Enterprise AI Architect Learning Path v3 - Overview

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Program Mission

Transform an experienced enterprise software architect into an Enterprise Agentic AI Architect through the design, implementation, operation, and governance of production-oriented AI systems.

This program leverages three interconnected portfolio projects:

1. RPG Catalog RAG Platform
2. Virtual GM Agent Platform
3. AI Publishing Company Platform

These projects provide practical experience with:

- Azure AI Foundry
- Azure OpenAI
- Azure AI Search
- Semantic Kernel
- LangGraph
- Model Context Protocol (MCP)
- LLMOps
- OpenTelemetry
- Security & Governance

The objective is not simply learning AI development but demonstrating enterprise architecture competency through working systems, architectural artifacts, evaluation frameworks, and governance documentation. [<File>003...pdf</File> | PDF], [002 Enterp...y Syllabus | PDF], [Technical...g Appendix | PDF], [Enterprise...rator_Plan | Word]

Candidate Profile

Designed for:

- Senior full-stack .NET developer
- 35+ years software experience
- Microsoft AI-900 certified
- Actively developing Python proficiency
- Interested in agentic AI architecture
- Building real-world portfolio projects

Program Outcomes

By completion, the learner will be able to:

- Design enterprise RAG architectures
- Build and evaluate retrieval systems
- Create autonomous and human-governed agents
- Develop MCP tool ecosystems
- Instrument AI systems with OpenTelemetry
- Apply LLMOps practices
- Conduct AI threat modeling
- Apply NIST AI RMF governance controls
- Create enterprise architecture documentation
- Present a complete AI architecture portfolio

Resource Catalog

The Technical Resource Catalog (R0–R21) is maintained as a separate appendix and may be reused across future versions of the learning path. [001 Enterp...bus [Nate] | PDF]

Phase 1 — Python & AI Engineering Foundations

Objectives

Build enterprise-grade Python skills while establishing AI engineering fundamentals.

Technical Focus

- Python OOP
- AsyncIO
- APIs
- File Processing
- Testing
- Logging
- Configuration Management
- Data Pipelines

Deliverables

RPG Catalog Ingestor

Capabilities:

- Scan source files
- Normalize metadata
- Produce structured JSON

- Perform validation
- Generate testing artifacts

Portfolio Artifacts

- Repository structure
- ADRs
- Initial architecture diagram
- Testing strategy

Definition of Done

- Working ingestion pipeline
- Unit tests passing
- Structured logging
- Architecture documentation completed

Phase 2 — Azure AI Foundry & RAG Foundations

Objectives

Develop a cloud-native retrieval architecture.

Technical Focus

- Azure AI Foundry
- Azure OpenAI
- Azure AI Search
- Embeddings
- Chunking
- Vector Search
- Hybrid Search

Deliverables

RPG Catalog RAG v0.5

Capabilities:

- Document ingestion
- Embedding generation
- Search index
- Basic grounded retrieval

Portfolio Artifacts

- C4 Context Diagram
- C4 Container Diagram
- Resource Architecture Note

Definition of Done

- Retrieval pipeline operational
- Index populated
- Grounded responses generated
- Citation support established

Phase 3 — Production RAG & Evaluation

Objectives

Transform the retrieval system into a production-quality architecture.

Technical Focus

- Hybrid Retrieval
- Metadata Filtering
- Citation Contracts
- Prompt Versioning
- Evaluation Pipelines
- Golden Question Testing

Deliverables

RPG Catalog RAG v1

Capabilities:

- Hybrid search
- Metadata filtering
- Source citations
- Evaluation workflows

Portfolio Artifacts

- Evaluation Report
- ADRs
- Retrieval Design Documentation

Definition of Done

- 50+ Golden Questions

- Retrieval evaluation completed
- Citation validation operational

Phase 4 — Agent Orchestration

Objectives

Develop controlled, stateful agent architectures.

Technical Focus

Microsoft Stack

- Semantic Kernel
- Plugins
- Functions
- Agent Patterns

Stateful Orchestration

- LangGraph
- Persistence
- Human-in-the-loop
- Workflow Control

Deliverables

Virtual GM v1

Capabilities:

- Session State
- Rules Retrieval
- Encounter Generation
- Continuity Management

Portfolio Artifacts

- Framework Selection ADR
- State Machine Diagram
- Agent Architecture Documentation

Definition of Done

- Stateful workflow operation
- Memory persistence

- Human approval checkpoints

Phase 5 — MCP & Tool Integration

Objectives

Create reusable AI tool infrastructure.

Technical Focus

- MCP Architecture
- JSON-RPC
- Tool Contracts
- Security Boundaries
- Permission Models

Deliverables

RPG MCP Server

Capabilities:

- Tool Discovery
- Rules Lookup
- Structured Tool Definitions

Portfolio Artifacts

- Tool Permission Matrix
- MCP Architecture Diagram
- Audit Strategy

Definition of Done

- MCP server operational
- Tools callable by agents
- Security controls documented

Phase 6 — AI Publishing Company Platform

Objectives

Build long-running collaborative agent systems.

Technical Focus

- Multi-Agent Design
- Editorial Workflows
- Human Governance
- State Persistence
- Content Operations

Deliverables

Publishing Agents v1

Agent Roles:

- Acquisition
- Developmental Editor
- Continuity Reviewer
- Copy Editor
- Publishing Coordinator

Portfolio Artifacts

- Workflow Diagrams
- Role Definitions
- Governance Notes

Definition of Done

- Idea → Outline → Draft → Review workflow
- Version tracking
- Approval checkpoints

Phase 7 — LLMOps, Observability & Deployment

Objectives

Make all systems observable, measurable, and deployable.

Technical Focus

- OpenTelemetry
- Application Insights
- Evaluation Automation
- Cost Tracking
- CI/CD
- Deployment Architecture

Deliverables

Instrumented AI Platform

Capabilities:

- Distributed Tracing
- Cost Monitoring
- Automated Evaluations
- Deployment Pipelines

Portfolio Artifacts

- Ops Dashboard
- Evaluation Pipeline
- Deployment Documentation

Definition of Done

- End-to-end tracing
- Regression testing
- Cost visibility
- Deployable architecture

Phase 8 — Security, Governance & Portfolio Completion

Objectives

Apply enterprise governance and security practices.

Technical Focus

Security

- OWASP Top 10 for LLMs
- Prompt Injection
- Tool Abuse
- Data Leakage

Governance

- NIST AI RMF
- Risk Registers
- AI Safety Processes
- Compliance Documentation

Deliverables

Enterprise Architecture Portfolio

Contents:

- Context Diagrams
- Container Diagrams
- Component Diagrams
- Sequence Diagrams
- Risk Register
- Governance Checklist
- Threat Models
- Evaluation Reports

Definition of Done

- Security reviews completed
- Governance package completed
- Portfolio ready for presentation

Certification Alignment

Recommended

Microsoft Applied Skills

- Azure AI Search
- Semantic Kernel

Microsoft AI-102

Target Window:

- End of Phase 3 through Phase 7

Certifications support the roadmap but do not gate progression.

Enterprise Translation Layer

Each project directly maps to enterprise use cases:

	Portfolio Project	Enterprise Equivalent
1	RPG Catalog RAG	Legal / Policy RAG
2	Virtual GM	Operations Agent
3	Publishing Company	Multi-Agent Business Workflow
4	MCP Server	Enterprise Tool Hub
5	Evaluation Suite	AI Quality Program
6	Governance Package	Enterprise AI Governance Framework

Graduation Criteria

The learner graduates when they can:

- Explain and justify architecture decisions
- Demonstrate working RAG systems
- Demonstrate stateful agents
- Implement secure MCP tools
- Measure AI quality through evaluations
- Operate AI systems with observability
- Produce enterprise governance artifacts
- Present a complete AI architecture portfolio

Final Outcome: Enterprise Agentic AI Architect Portfolio.